Relevant passage ranking

Alpha Follows the Hops

Query Complexity and Score Fusion in Graph-Augmented Re-ranking

Tanish Verma Narmety Vamshi Pruthviraj Sharad Salunke

Mehta Family School of Data Science & Artificial Intelligence, IIT Roorkee

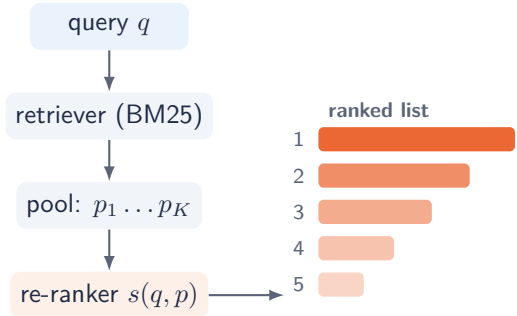


Part 1

Background

The task: rank passages for a query

- Input: a query q and a pool of candidate passages $p_1, \ldots, p_K$.
- Output: the passages ordered by relevance to q.
- A **scoring function** gives each pair (q, p_i) a score $s(q, p_i)$; the ranking sorts by it.
- Usual pipeline: a cheap retriever fills the pool (BM25, top-K), an expensive model *re-ranks* it.
- Evaluated with nDCG@10 and MRR@10.



Scoring functions: one score per pair

TF-IDF

$\sum_{t \in q} \mathrm{tf}(t, p) \cdot \mathrm{idf}(t)$

Exact term overlap, weighted by rarity.

BM25

$\sum_{t \in q} \mathrm{idf}(t) \, \dfrac{\mathrm{tf}\,(k_1 + 1)}{\mathrm{tf} + k_1 B_p}$

TF saturation and length normalisation; the standard retriever.

Bi-encoder

$\cos\big(E(q),\, E(p)\big)$

Query and passage encoded separately; fast, indexable.

Cross-encoder

$\mathrm{BERT}\big([q\,;\,p]\big)$

Joint encoding; most accurate, too slow to index.

lexical $\longrightarrow$ neural: more accurate, more expensive

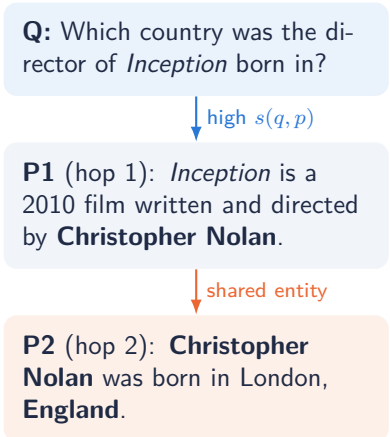
All of them score each passage **on its own**: $s(q, p_i)$ never looks at the other candidates.

Where it breaks: multi-hop questions

Single-hop: one passage answers the query.

Multi-hop: the answer needs a **chain** of passages.

- The second passage (about the director) shares almost no words with the query.
- A pointwise scorer ranks it low, although it is needed.
- Its relevance shows only *through its link* to P1.



Illustrative example.

GAR: Graph-based Adaptive Re-ranking

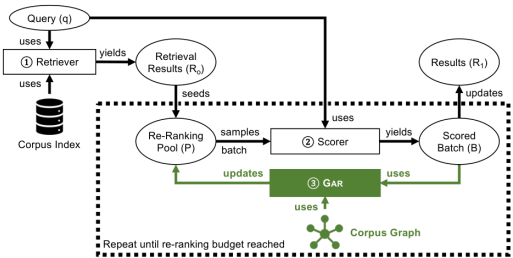


Figure 1: Overview of GAR. Traditional re-ranking exclusively scores results seeded by the retriever. GAR (in green) adapts the re-ranking pool after each batch based on the computed scores and a pre-computed graph of the corpus.

Screenshot from MacAvaney et al. (CIKM 2022).

- **Clustering hypothesis**: passages close to a relevant passage tend to be relevant too.
- Offline: a **corpus graph**, each passage linked to its k most similar passages.
- Online: after each scored batch, the *neighbours* of the top passages join the pool.
- Fixes the recall limit of the first-stage pool; the scorer itself stays pointwise.

GAR: the re-ranking loop

Algorithm 1 Graph-based Adaptive Re-Ranking

Input: Initial ranking R_0, batch size b, budget c, corpus graph G
Output: Re-Ranked pool R_1

$R_1 \leftarrow \emptyset$ ▷ Re-Ranking results
$P \leftarrow R_0$ ▷ Re-ranking pool
$F \leftarrow \emptyset$ ▷ Graph frontier
do
  $B \leftarrow \textsc{Score}(\text{top } b \text{ from } P, \text{subject to } c)$ ▷ e.g., monoT5
  $R_1 \leftarrow R_1 \cup B$ ▷ Add batch to results
  $R_0 \leftarrow R_0 \setminus B$ ▷ Discard batch from initial ranking
  $F \leftarrow F \setminus B$ ▷ Discard batch from frontier
  $F \leftarrow F \cup (\textsc{Neighbours}(B, G) \setminus R_1)$ ▷ Update frontier
  $P \leftarrow \begin{cases} R_0 & \text{if } P = F \\ F & \text{if } P = R_0 \end{cases}$ ▷ Alternate initial ranking and frontier
while $|R_1| < c$
$R_1 \leftarrow R_1 \cup \textsc{Backfill}(R_0, R_1)$ ▷ Backfill remaining items

Algorithm 1 of MacAvaney et al. (CIKM 2022).

- Alternates between the initial ranking R_0 and the *graph frontier* F.
- Up to $+8\%$ nDCG for BM25 $\gg$ monoT5 on MS MARCO.
- The graph decides **which** passages get scored, not **how** they are scored.

Next step: let the graph change the *score* itself, so a passage is judged together with its neighbours $\Rightarrow$ ReGAT-BERT.

ReGAT-BERT: the idea

Semantic score s_{sem}

A BERT cross-encoder reads (q, p) jointly.
Strong on *direct* matches.

Contextual score s_{ctx}

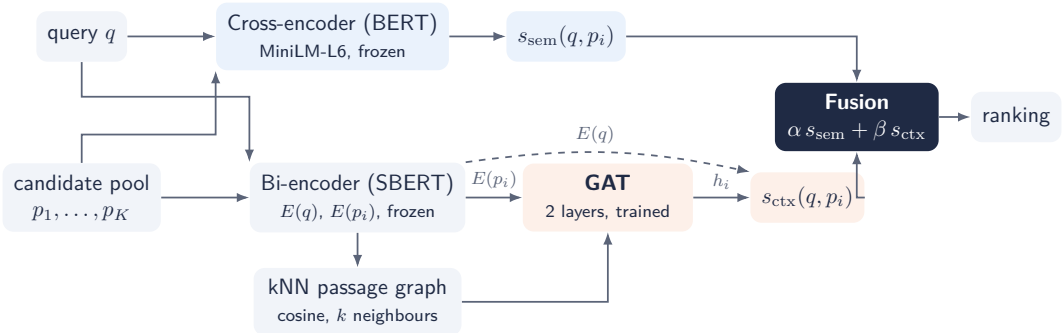
Passages form a graph; a Graph Attention Network refines each passage embedding with its neighbours.
Scores the passage *in context*.

- Two complementary views of relevance.
- Final score = a weighted sum of the two:

$$\text{score}(q, p) = \alpha\, s_{\text{sem}} + \beta\, s_{\text{ctx}}$$

- α, β are learned **once**, then the **same weights apply to every query**.

ReGAT-BERT architecture

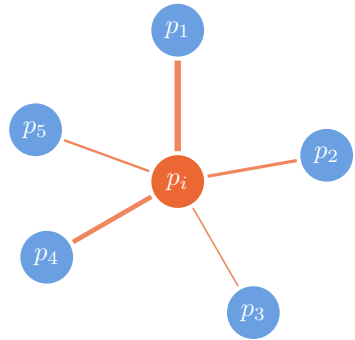


Our diagram of the pipeline as reproduced in this project (models as used in our experiments).

GAT step 1: build a graph over the pool

One graph per query, over its K candidate passages.

node i	candidate passage p_i ($K = 100$ on MS MARCO, 10–20 in closed pools)
$x_i = E(p_i) \in \mathbb{R}^d$	node feature: the frozen bi-encoder embedding of p_i, $d = 384$
edge $j \to i$	p_j is one of the k nearest neighbours of p_i by cosine, and $\cos(x_i, x_j) \geq 0.3$; $k = 8$ (MS MARCO), 3 (multi-hop)
self-loop	edge $i \to i$, so a passage keeps its own information
$\mathcal{N}(i)$	neighbourhood of i: its neighbours and itself
X, A	all node features $[x_1, \ldots, x_K]$ and the adjacency matrix ($A_{ij} = 1$ if edge)



edge width $\propto$ attention π_{ij} (step 2): the GAT learns which neighbours matter

GAT step 2: how much should p_i listen to p_j?

Attention score (GATv2, one head):

$$e_{ij} = \mathbf{a}^\top \,\mathrm{LeakyReLU}\big(\mathbf{W}_t\, x_i + \mathbf{W}_s\, x_j\big)$$

Attention weight: softmax over the neighbourhood

$$\pi_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})}$$

$\pi_{ij} \geq 0$, $\sum_j \pi_{ij} = 1$.

Called α_{ij} in the GAT papers; renamed to avoid a clash with the fusion weight α.

x_i, x_j	embedding of the passage being updated (i) and of a neighbour (j)
$\mathbf{W}_t, \mathbf{W}_s$	learned $d' \times d$ projections for the target (i) and source (j) node
LeakyReLU	$\max(z, 0.2z)$ per coordinate (non-linearity)
$\mathbf{a} \in \mathbb{R}^{d'}$	learned attention vector; $\mathbf{a}^\top(\cdot)$ is a dot product: vector $\rightarrow$ *one number*
e_{ij}	raw importance of neighbour j for i
π_{ij}	normalised attention weight

GATv2 applies $\mathbf{a}$ *after* the non-linearity ("dynamic attention").

GAT step 3: aggregate, multi-head, two layers

One head: weighted average of projected neighbours

$$x'_i = \sum_{j \in \mathcal{N}(i)} \pi_{ij} \, \mathbf{W}_s \, x_j + \mathbf{b}$$

Layer 1: $M = 4$ heads, outputs concatenated

$$h_i^{(1)} = \Big\|_{m=1}^{4} \Big(\sum_j \pi_{ij}^{(m)} \mathbf{W}_s^{(m)} x_j + \mathbf{b}^{(m)} \Big) \in \mathbb{R}^{4 \cdot 128}$$

Layer 2: one head, back to dimension d

$$h_i^{(2)} = \sum_j \pi_{ij}^{(2)} \, \mathbf{W}_s^{(2)} \, \mathrm{ELU}\big(h_j^{(1)}\big) + \mathbf{b}^{(2)} \in \mathbb{R}^{384}$$

$\mathbf{b}$	learned bias vector
m	head index; each head has its own $\mathbf{W}$, $\mathbf{a}$, so it can attend to a different kind of relation
$\|$	concatenation of the 4 head outputs ($4 \times 128 = 512$ numbers)
$\mathrm{ELU}(z)$	z if $z > 0$, else $e^z - 1$ (smooth non-linearity between layers)
dropout	0.1 on the attention weights while training

2 layers = 2 graph hops: p_i also hears its neighbours' neighbours.

GAT step 4: refined passage → contextual score

Residual: keep the meaning, add a correction

$$z_i = x_i + \mathrm{GAT}(X, A)_i$$

LayerNorm, then unit length

$$\mathrm{LN}(z) = \boldsymbol{\gamma} \odot \frac{z - \mu_z}{\sqrt{\sigma_z^2 + \epsilon}} + \boldsymbol{\eta}$$

$$h_i = \mathrm{LN}(z_i) \,/\, \|\mathrm{LN}(z_i)\|_2$$

Contextual score

$$s_{\mathrm{ctx}}(q, p_i) = E(q)^\top h_i = \cos\big(E(q), h_i\big)$$

$\mathrm{GAT}(X, A)_i$	2-layer GAT output $h_i^{(2)}$ for node i, given all features X and the graph A
LN	Layer Normalisation: re-centres and re-scales one vector over its own d coordinates
μ_z, σ_z	mean and std of the d coordinates of z
$\boldsymbol{\gamma}$, $\boldsymbol{\eta}$	learned scale and shift ($\in \mathbb{R}^d$); $\odot$ = element-wise product
$\|\cdot\|_2$	Euclidean length: h_i becomes a unit vector
$E(q)$	unit-length query embedding from the *same* bi-encoder

h_i and $E(q)$ share one $\mathbb{R}^{384}$ space: the residual starts h_i at the bi-encoder vector x_i.

Training the GAT: InfoNCE loss

$$\mathcal{L}_{\mathrm{GAT}} = -\log \frac{\sum_{p^+} \exp\left(s_{\mathrm{ctx}}(q, p^+)/\tau\right)}{\sum_{j=1}^{K} \exp\left(s_{\mathrm{ctx}}(q, p_j)/\tau\right)}$$

- The fraction is the share of the softmax over the pool that falls on the gold passages.
- Minimising $\mathcal{L}$ pulls gold h_i towards $E(q)$ and pushes the other pool passages away.
- Trained: $\mathbf{W}_s, \mathbf{W}_t, \mathbf{a}, \mathbf{b}, \boldsymbol{\gamma}, \boldsymbol{\eta}$. Bi-encoder and cross-encoder stay **frozen**.
- AdamW, lr 10^{-4}, early stopping on val nDCG@10; trained **once** per dataset, then frozen for all fusion experiments.

InfoNCE	a contrastive loss: cross-entropy of picking the gold passages out of the pool
p^+	gold (relevant) passages of q in the pool
p_j, K	all pool passages, pool size
$\tau = 0.05$	temperature: dividing by a small τ sharpens the softmax, so small score gaps matter
exp, log	natural exponential and logarithm

ReGAT-BERT fusion strategy

$$\text{score}(q,p) = \alpha \cdot s_{\text{sem}}(q,p) + \beta \cdot s_{\text{ctx}}(q,p)$$

α, β: two free scalars, learned with a pointwise **BCE** loss:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{(q,p)} \Big[y \log \sigma(\text{score}) + (1-y) \log \big(1 - \sigma(\text{score})\big) \Big]$$

BCE	Binary Cross-Entropy: penalises a confident wrong relevance probability
$y \in \{0,1\}$	relevance label of the pair (q,p)
$\sigma(z)$	sigmoid $1/(1+e^{-z})$: score $\to$ probability
N	number of training pairs, each judged on its own

- Raw scores are fused as they are: the cross-encoder logit and the cosine live on **different scales**.
- One pair (α, β) for **all queries**: a keyword query and a 4-hop question get the same mix.

Part 2

Our study

Problem statement

Should the weight between the semantic and the contextual score be **the same for every query**, or should it depend on the query and its passages?

Simple / single-hop queries

One passage answers it; the cross-encoder sees everything it needs.

Expect $\alpha > 0.5$ (trust s_{sem}).

Complex / multi-hop queries

Evidence is spread over linked passages; only the graph sees the links.

Expect $\alpha < 0.5$ (trust s_{ctx}).

Hypothesis: α should *fall as the number of hops rises*, so a query-dependent $\alpha(\cdot)$ should beat a single fixed α.

Our score and the normalisers

$$\text{score}(q, p_i) = \alpha \cdot \tilde{s}_{\text{sem}}(q, p_i) + (1 - \alpha) \cdot \tilde{s}_{\text{ctx}}(q, p_i), \qquad \alpha \in (0, 1)$$

$\tilde{s}$ = score normalised *within each query's pool*, so α is a true mixing weight. $\alpha = 1$: cross-encoder only; $\alpha = 0$: GAT only.

None

$\tilde{s}_i = s_i$

Raw scores (paper setting).

Min–max

$\tilde{s}_i = \frac{s_i - \min s}{\max s - \min s}$

Maps to $[0, 1]$; our primary normaliser.

Z-score

$\tilde{s}_i = \frac{s_i - \mu_q}{\sigma_q}$

Zero mean, unit variance per query.

Rank

$\tilde{s}_i = 1 - \frac{\text{rank}_i - 1}{K - 1}$

Only the order counts; scale-free.

s_i: raw score of the i-th passage in the query's pool. $\min s$, $\max s$, μ_q, σ_q: minimum, maximum, mean and standard deviation over that pool. rank_i: position of p_i when the pool is sorted by s ($1 =$ best). K: pool size.

Modelling α: the gate network

$$\alpha(x) = \epsilon + (1 - 2\epsilon) \cdot \sigma\Big(\mathbf{w}_2^\top \operatorname{ReLU}\big(\mathbf{W}_1 x + \mathbf{b}_1\big) + b_2\Big)$$

Gate	input x (what α may depend on)
fixed	none: one learned scalar for all queries
Q	query embedding $E(q) \in \mathbb{R}^{384}$
S	12 score statistics $f(s_{\text{sem}}, s_{\text{ctx}})$
QS	$[E(q) \,\|\, f]$
QS+C	$[E(q) \,\|\, f \,\|\, c]$, c = length, #entities

Every query gets its **own** α, computed from x at test time; no labels are used.

x	gate input for this query (left)
$\mathbf{W}_1, \mathbf{b}_1$	hidden layer, 64 units (dropout 0.1)
$\operatorname{ReLU}(z)$	$\max(z, 0)$
$\mathbf{w}_2, b_2$	output layer: one number (a logit)
$\sigma(z)$	sigmoid, maps the logit into $(0, 1)$
$\epsilon = 10^{-6}$	keeps α strictly inside $(0, 1)$
$\|$	concatenation of input vectors

Gate inputs and training

S features (12 per query, from raw scores):

For each of s_{sem} and s_{ctx} over the pool:

mean, std	level and spread of the scores
top-1	best score
top-1 − top-2	margin: how confident the scorer is
entropy	of softmax(scores): how peaked

plus Spearman $\rho(s_{\text{sem}}, s_{\text{ctx}})$ (do the two rankings agree?) and the overlap of their top-10.

C: query length in words and number of named entities (spaCy).

Listwise loss with a pull towards α_0:

$$\mathcal{L} = -\log \sum_{p^+} \text{softmax}_i\left(\frac{\text{score}_i}{T}\right) + \lambda\,(\alpha - \alpha_0)^2$$

score_i	fused score of pool passage i
softmax_i	over the query's pool; first term = probability mass on gold p^+
T	learned temperature
α_0	best single fixed α on the validation grid (step 0.05)
$\lambda = 0.01$	strength of the pull towards α_0

$b_2 = \text{logit}(\alpha_0)$, $\mathbf{w}_2 \approx 0$ at start: every gate starts as the fixed baseline. AdamW, early stopping on val, 5 seeds. CE and GAT frozen.

Evaluation metric: nDCG@k

$$\mathrm{DCG@}k = \sum_{i=1}^{k} \frac{\mathrm{rel}_i}{\log_2(i+1)} \qquad \mathrm{nDCG@}k = \frac{\mathrm{DCG@}k}{\mathrm{IDCG@}k} \in [0,1]$$

rel_i	label of the passage at rank i: 1 = gold, 0 = not (graded on TREC DL)
$\log_2(i+1)$	position discount: rank 1 counts fully, rank 3 half, rank 10 only 0.29
IDCG@k	DCG@k of the ideal ranking (all gold passages on top)
k	cut-off: only the top k passages count

- Reported $\times 100$; credits **every** gold passage and how high it sits, key when a question has several.
- **nDCG@10** is our primary metric; α, gates and normalisers are chosen on validation nDCG@10.
- Computed with ir_measures (trec_eval convention, linear gain).

Example: 2-hop question, 2 gold passages (■), $k = 5$.

ranking A	**1**	2	**3**	4	5
ranking B	1	**2**	**3**	4	5
ideal	**1**	**2**	3	4	5

$\mathrm{IDCG} = 1 + \frac{1}{\log_2 3} = 1.63$

A: $(1 + \frac{1}{2})/1.63 = \mathbf{0.92}$

B: $(0.63 + \frac{1}{2})/1.63 = \mathbf{0.69}$

Both have Recall@5 = 1; nDCG prefers A, which puts a gold passage first.

Datasets: from single-hop to multi-hop

MS MARCO
single-hop
Real Bing queries; 1 relevant passage per query.

HotpotQA
2-hop
Bridge and comparison questions.

BEIR-HQA
2-hop, open
HotpotQA queries over 5M Wikipedia passages.

2Wiki
2–4 hops
4 types: comparison, compositional, inference, bridge-comparison.

MuSiQue
2, 3, 4 hops
Built by composing single-hop questions; hops labelled.

Dataset	Setting	K	GAT train	Fusion train	Val	Test	Gold/pool	Complexity labels
MS MARCO (A)	open (BM25)	100	30,000	10,000	2,000	6,980	–	proxies only
MS MARCO (B)	open (BM25)	100	1,870	1,150	698	1,745	–	proxies only
MuSiQue	closed pool	20	10,000	7,000	2,938	2,417	2.65	hops 2/3/4
HotpotQA	closed pool	10	20,000	10,000	2,000	7,405	2.00	bridge/comparison
BEIR-HotpotQA	open (BM25)	100	20,000	10,000	2,000	7,405	1.59	bridge/comparison
2Wiki	closed pool	10	20,000	10,000	2,000	12,576	2.44	4 types

Table 1: Datasets and splits (queries). Val is always carved from the training data; test is the official dev set (MS MARCO A: dev/small; MuSiQue, HotpotQA, 2Wiki: dev; BEIR: test). MS MARCO B splits dev/small itself, so the cross-encoder never saw any of its queries. Gold/pool = mean number of relevant passages among the K test candidates.

Part 3

Results

1. The best α differs by dataset

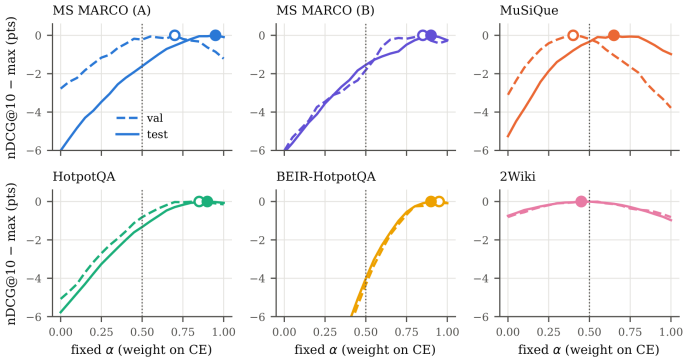


Figure 1: nDCG@10 as a function of the fixed weight α (min–max), relative to each curve's maximum; dashed = validation, solid = test, circles = arg-max. Left of the dotted line the contextual score has more weight. The single-hop reference (MS MARCO) peaks far to the right; MuSiQue (validation) and 2Wiki peak left of 0.5. The y-axis is clipped at -6 points.

- Each curve: nDCG@10 of the fixed fusion as α goes from 0 (GAT only) to 1 (CE only).
- Single-hop **MS MARCO** peaks far right: $\alpha = 0.70$–0.95.
- Multi-hop **MuSiQue** (val, $\alpha = 0.40$) and **2Wiki** ($\alpha = 0.45$) peak **left of 0.5**: the contextual score gets more weight.

1. More hops ⇒ lower α (MuSiQue)

		nDCG@10 of each signal alone				Best fixed α						% prefer ctx	
		val		test		val			test				
Hops	n (test)	CE	GAT	CE	GAT	mm	z	rank	mm	z	rank	val	test
2-hop	1,252	76.9	76.8	73.8	68.0	**0.40**	0.50	**0.45**	0.80	0.80	0.95	48	40
3-hop	760	69.2	71.0	71.8	66.7	**0.40**	**0.40**	**0.30**	0.55	0.65	0.85	54	39
4-hop	405	66.1	71.7	61.3	63.3	**0.25**	**0.30**	**0.00**	**0.40**	**0.40**	**0.00**	60	52

Table 3: MuSiQue by hop count. Left: the cross-encoder loses 10.8 (val) and 12.5 (test) nDCG@10 points from 2 to 4 hops, the GAT only 5.1 and 4.7, and overtakes it on 4-hop questions. Middle: the best fixed α falls with the hop count under every scale-free normaliser (mm = min–max, z = z-score); bold = below 0.5, i.e. the contextual score dominates. Right: share of α-sensitive questions whose optimal α-interval is centred below 0.5. Tie-robust per-question test of the trend: Spearman(preferred α, hops) = -0.072 [-0.110, -0.032] on val ($p = 0.0005$) and -0.068 [-0.110, -0.025] on test ($p = 0.004$).

Best α falls with each hop. Test, min–max: **0.80** → **0.55** → **0.40**; val: 0.40 → 0.40 → 0.25.

Why: 2 → 4 hops, the cross-encoder drops 12.5 points (test), the GAT only 4.7.

1. The trend holds under every normaliser

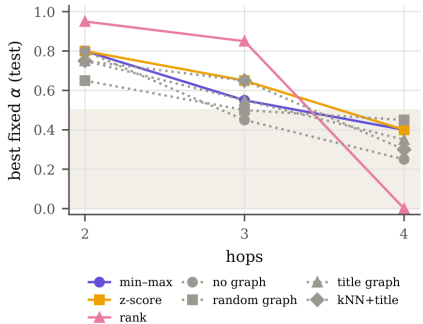


Figure 3: MuSiQue: best fixed α on test per hop count under the three scale-free normalisers (coloured) and for GATs retrained on four other graphs (grey, dotted; min–max). Shaded: $\alpha < 0.5$. Every setting declines from 2 to 4 hops, and every 4-hop optimum is below 0.5.

- Min–max, z-score and rank all give a **monotone decline** of the best α from 2 to 4 hops.
- 4-hop questions sit **below 0.5** in all 14 normaliser / graph / split settings.
- Per question: preferred α correlates negatively with hop count (Spearman $\rho \approx -0.07$, $p \leq 0.004$ on val and test).

1. Same pattern by question type (2Wiki)

	nDCG@10 (test)		Best fixed α		% prefer	Learned Q gate: mean α and ΔnDCG@10 vs. fixed (test)					
Type	CE	GAT	val	test	ctx	min–max		z-score		rank	
comparison	97.3	95.5	0.90	0.65	30	0.72	+0.10	0.52	-0.04	0.59	-0.07
compositional	88.5	88.9	**0.45**	0.50	53	0.45	-0.01	0.34	-0.08	0.44	+0.25$^{\blacktriangle}$
inference	86.7	87.6	**0.25**	**0.45**	59	0.30	-0.23$^{\triangledown}$	0.22	-0.03	0.38	+0.27
bridge-comparison	90.1	92.0	**0.25**	**0.20**	67	0.35	+0.16$^{\blacktriangle}$	0.33	+0.22$^{\blacktriangle}$	0.17	+0.81$^{\blacktriangle}$

Table 4: 2WikiMultiHopQA by question type, ordered from the simplest (comparison: both entities named in the question) to the most compositional (bridge-comparison). Bold $\alpha < 0.5$. The query-only gate, which never sees the type label, gives comparison questions the highest α under all three normalisers, and gains significantly on bridge-comparison questions ($^{\blacktriangle}$/$^{\triangledown}$: paired bootstrap $p < 0.05$).

Comparison:
$\alpha = 0.65$–0.90,
cross-encoder wins.

Inference, bridge-comp.:
$\alpha = 0.20$–0.45,
GAT wins.

Q gate, blind to the type:
highest α on comparison,
gains on bridge-comp.

2. Adaptive α gains on complex queries

Norm.	Gate	MSM-A	MSM-B	MuSiQue	HotpotQA	BEIR-HQA	2Wiki
none	Q	-0.48$^{\triangledown}$	+0.01	+0.00	-0.00	-0.00	+0.00
	QS	-0.48$^{\triangledown}$	+0.00	+0.00	+0.01	-0.00$^{\triangledown}$	+0.00
	S	-0.49$^{\triangledown}$	+0.00	+0.02	+0.01	-0.00	-0.00
minmax	Q	**+0.00**$^{\blacktriangle}$	+0.00	+0.00	-0.02	-0.00	+0.03
	QS	+0.00	+0.01	-0.01	+0.02	+0.00	-0.12$^{\triangledown}$
	S	+0.00	**+0.02**$^{\blacktriangle}$	-0.00	+0.04	+0.01	+0.00
	QS+C	–	–	+0.00	+0.03	-0.00	-0.11$^{\triangledown}$
zscore	Q	+0.01	**+0.05**$^{\blacktriangle}$	-0.15$^{\triangledown}$	+0.06	**+0.18**$^{\blacktriangle}$	+0.00
	QS	**+0.02**$^{\blacktriangle}$	-0.01	-0.14$^{\triangledown}$	+0.00	-0.01$^{\triangledown}$	-0.00
	S	+0.01	+0.01	-0.12$^{\triangledown}$	+0.01	-0.01	-0.01$^{\triangledown}$
rank	Q	+0.01	+0.02	**+0.43**$^{\blacktriangle}$	-0.01	-0.01	**+0.30**$^{\blacktriangle}$
	QS	-0.12	-0.04	**+1.22**$^{\blacktriangle}$	-0.04	-0.04$^{\triangledown}$	**+0.41**$^{\blacktriangle}$
	S	-0.10	-0.02	**+1.37**$^{\blacktriangle}$	-0.00	-0.02$^{\triangledown}$	**+0.19**$^{\blacktriangle}$

Table 6: Every adaptive gate against the best fixed α under the same normaliser: test ΔnDCG@10 $\times 100$ (paired bootstrap; $^{\blacktriangle}$ significantly better, $^{\triangledown}$ worse, $p < 0.05$ uncorrected).

7 of the 11 significant gains are on **multi-hop** sets; largest on MuSiQue (rank): +0.43 / +1.22 / +1.37.

MS MARCO gains ≤ 0.05. Oracle per-query α headroom: 7.4 (MuSiQue) vs. 5.0 (MS MARCO).

2. Transfer: α follows the target data

			Transferred gate, test nDCG@10				In-domain		Q gate
Setting	Train	Test	fixed	Q	QS	S	fixed	α_0	$\bar{\alpha}$ (sd)
leave-one-out	MSM+HQA	MuSiQue	71.99	71.93	71.98	72.00	71.28	0.70	0.73 (0.02)
mixed	MSM+MuSiQue+HQA	MuSiQue	71.98	71.77	71.81	71.79	71.28	0.55	0.54 (0.16)
zero-shot	MSM	BEIR-HotpotQA	71.45	71.45	71.45	71.46	72.63	0.70	0.70 (0.00)
zero-shot	MSM	HotpotQA	89.24	89.24	89.25	89.20	89.89	0.70	0.70 (0.00)
zero-shot	MSM	2Wiki	90.99	90.99	90.99	90.99	91.75	0.70	0.70 (0.00)

Table 9: Transfer and mixed training of the gate (E4, min–max, $\times 100$, 5 seeds). Zero-shot uses the frozen MS MARCO GAT for s_{ctx}; mixed and leave-one-out use each dataset's own GAT. α_0 is the fixed α the gate starts from (selected on the training sources' val).

One gate on mixed data (MS MARCO + MuSiQue + HotpotQA) gives MuSiQue a mean α of 0.54, well below the 0.70 of an MS MARCO-only gate, and varies it per query (sd 0.16).

Zero-shot MS MARCO gates keep $\alpha = 0.70$ and lose to the target's own setting, e.g. 2Wiki 90.99 vs. 91.75.

Conclusion

- **Alpha follows the hops**: the more reasoning steps a query needs, the more weight belongs on the graph-contextual score.
- **MuSiQue** (hops): best α 0.80 $\rightarrow$ 0.55 $\rightarrow$ 0.40 for 2 $\rightarrow$ 3 $\rightarrow$ 4 hops (test); 4-hop below 0.5 in every setting.
- **2Wiki** (types): best α 0.65 $\rightarrow$ 0.50 $\rightarrow$ 0.45 $\rightarrow$ 0.20 for comparison $\rightarrow$ compositional $\rightarrow$ inference $\rightarrow$ bridge-comparison (test; val 0.90 $\rightarrow$ 0.45 $\rightarrow$ 0.25 $\rightarrow$ 0.25).
- Mechanism: the cross-encoder degrades with hops much faster than the GAT.
- One fixed weight is a compromise; query-dependent gates gain mostly on multi-hop data.

Limits

HotpotQA keeps a high α (single-hop shortcuts). Min–max gates mostly stay at their start value.

Next

Gates with explicit complexity signals (predicted hop count), and fusing at the passage-pair level.

Thank you

Questions?

MacAvaney, Tonellotto, Macdonald. Adaptive Re-Ranking with a Corpus Graph. CIKM 2022.
Brody, Alon, Yahav. How Attentive are Graph Attention Networks? ICLR 2022. Trivedi et al. MuSiQue. TACL 2022.

Yang et al. HotpotQA. EMNLP 2018. Ho et al. 2WikiMultiHopQA. COLING 2020. Nguyen et al. MS MARCO. 2016. Thakur et al. BEIR. NeurIPS 2021.